EngSci 721

Inverse Problems and Learning From Data

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1. *Basic concepts* [5 lectures + 1 Tutorial]

   Forward vs inverse problems. Well-posed vs ill-posed problems. Algebra and calculus of inverse problems (left and right inverses, generalised and pseudo inverses, resolution operators, matrix calculus). Representing higher dimensional problems (image data etc).

2. *Instability and regularisation in linear and nonlinear problems* [6 lectures + 1 Tutorial]

   Instability and related issues for generalised inverses. Introduction to regularisation and trade-offs. Tikhonov regularisation. Higher-order Tikhonov regularisation. Sparsity and regularisation using different norms. Truncated singular value decomposition. Iterative regularisation, including stochastic/mini-batch gradient descent.

3. *Further topics* [3 lectures + 1 Tutorial]

   Regularisation parameter choice, including statistical and machine learning views of regularisation. Confidence sets for linear and nonlinear models. Physics-informed machine learning and neural networks.

Module overview

Inverse Problems and Learning From Data (*Oliver Maclaren*)
[~14 lectures/3 tutorials]

Lectures 9–10: SVD and truncated SVD regularisation

Topics:

- Singular Value Decomposition
  - The ‘crown jewel’ of linear algebra!
  - Generalises eigenvalue analysis to general (non-square etc) matrices
- Truncated Singular Value Decomposition
  - As regularisation scheme
  - Relation to Tikhonov regularisation

Eng Sci 721 : Lecture 9

Regularisation in linear problems: Matrix factorisation & the (truncated) Singular Value Decomposition (SVD)

- Matrix factorisations in general
- SVD (type of matrix factorisation)
  - extension of eigen analysis
  - insight/calculation for inverses, resolution, effect of regularis.
  - the 'crown jewel' of linear algebra
- Truncated SVD.
  - as regularisation scheme.
  - connection to Tikhonov.

Bonus: rank factorisation & extension to nonlinear epi-mono factorisations

Matrix factorisations

Matrices play a central role in (linear) inverse problems, eg representing linear forward maps, multidimensional data, etc.

It is frequently useful to decompose or factorise a matrix A into two or more factors, eg

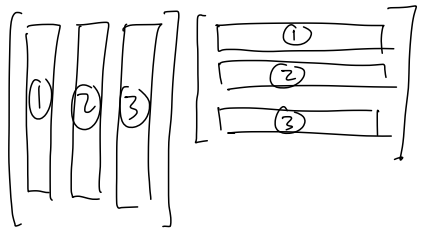
$$\boxed{A = BC} \text{ or } \boxed{A = BCD} \text{ etc.}$$

These are useful if B &/or C, D etc have particular structure that give us insight into A or allow us to operate with A, solve eq^{n}s involving A etc.

Outer products

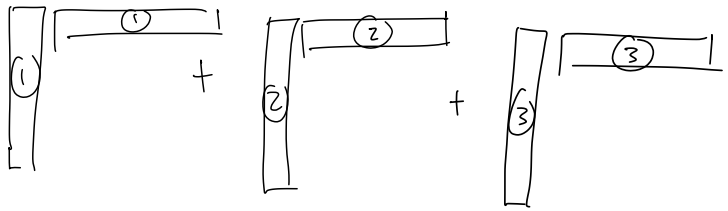
Recall that we can think of an arbitrary dimensionally consistent matrix multiplication AB in terms of outer products, eg

$AB =$



A B

$=$



$=$ sum of outer products:

$$\text{col}_i(A)\,\text{row}_i(B) = c_i(A)\,r_i^T(B)$$

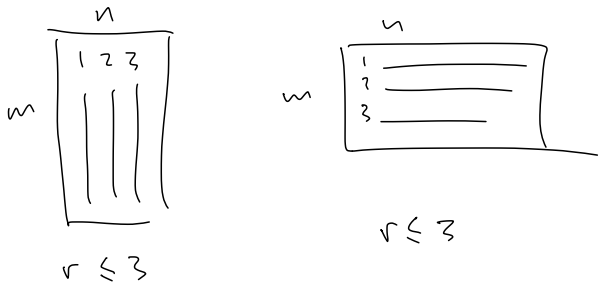
recall r^T & row convention.

Rank

The rank of a matrix is the number of linearly independent columns of the matrix

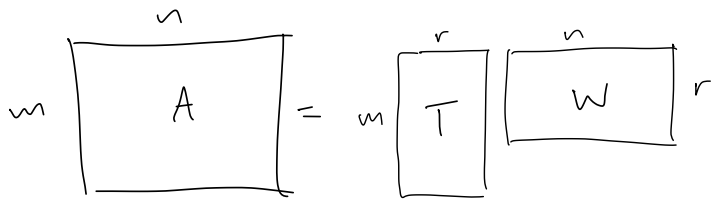
A fundamental theorem of linear algebra is that this is also equal to the number of linearly independent rows of the matrix

we have $\text{rank}(A) \leq \min\{m, n\}$ for an $m \times n$ matrix A



Rank factorisation

A rank factorisation for a matrix A of shape $m \times n$ & rank r is a factorisation into an $m \times r$ 'tall/square' matrix T & an $r \times n$ 'wide/square' matrix W:



$$\Rightarrow \boxed{A = \sum_{i=1}^{r} t_i w_i^T}$$ } sum of outer products

(w_i^T: rows of W)

↳ expansion form

→ It turns out these always exist

→ are not unique: many useful types exist

→

Simple example

Suppose A has linearly indep. col.

eg $A = \begin{bmatrix} ① & ② & ③ \end{bmatrix}$ & all L.I.

obv. $A = AI$

$$= \begin{bmatrix} ① & ② & ③ \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

T W

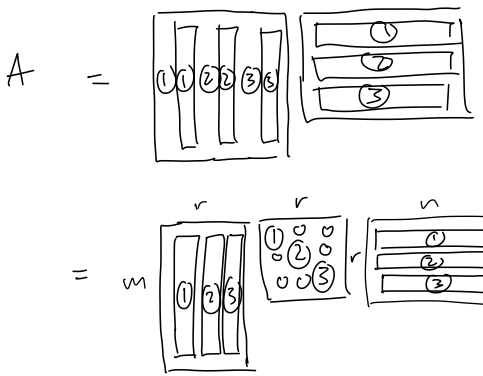
$$= \begin{bmatrix} ① & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & ② & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & ③ \end{bmatrix}$$

Suppose we rescale 'T' columns, $t_i = s_i \bar{t}_i$

get $$\boxed{A = \sum_{i=1}^{r} t_i w_i^T = \sum_{i=1}^{r} s_i \bar{t}_i w_i^T}$$

→

leads to eg



note:
- AS ← diag. = scale A cols by corresp. diag.
- diag → SA = scale rows of A by corresp. diag.

$\Rightarrow$ $A = TSW$ $\Leftrightarrow$ $A = \sum_i^r t_i s_i w_i^T$ ← expansion form

(T: 'tall', S: 'scale', W: 'wide'; $\sum_i^r t_i s_i w_i^T$: scalar, vector, vector)

More generally,

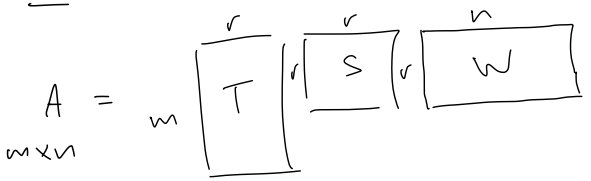
$A = TMW$ ← tho' no simple expansion version since M not diag.

('tall', 'mixing' (not diagonal, but invertible), 'wide')

[These ideas form foundation of
- singular value decomposition (SVD)
- CUR factorisation & matrix sketches.]

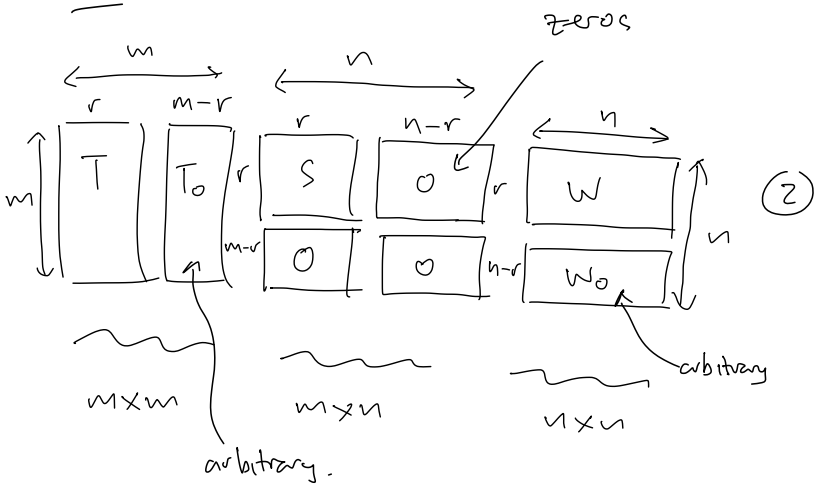
Note: extending/padding a rank factorisation

Given:



①

can write as:



②

arbitrary.

---

call ① 'reduced'

② 'full' or 'padded'

Side note:

We will now consider various forms of these matrix/linear factorizations

→ First though, note that we can actually extend concept of A = Tall-Wide to nonlinear f = mono ∘ epi ∘ !

monomorphism = 1-1 nonlinear function ~ 'tall, LI col' matrix

epimorphism = onto nonlinear function ~ 'wide, LI row' matrix

f: 

→ I haven't seen much on this tho'!

↳ research topic ??

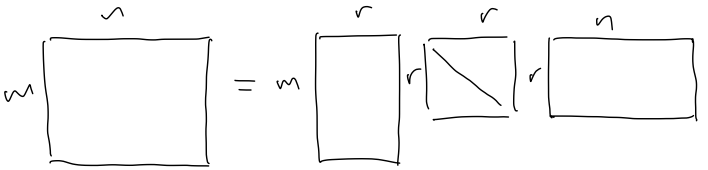
SVD (reduced)

The singular value decomposition (SVD) of an arbitrary $m \times n$ matrix A with rank r always exists and has the (reduced) form:

$$A = U_r \Lambda_r V_r^T$$

(some use Σ instead of Λ but we want sum!)

with shapes:



A arbitrary

U_r tall, cols are orthonormal & linearly indep.

Λ_r square & diagonal

V_r^T wide, rows are orthonormal & linearly indep.

expansion form:

$$A = \sum_{i=1}^{r} u_i \lambda_i v_i^T$$

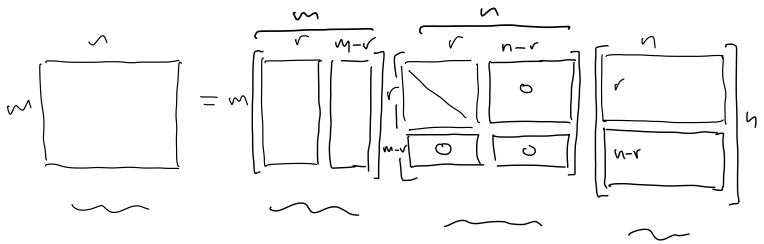
(r: number; u_i: col vector; v_i^T: row vector)

SVD (full - usually what people mean, tho' implicit.)

The 'full' SVD is a 'padded' version of the reduced SVD with form

$$A = U \Lambda V^T$$

with shapes:



A
arbitrary $m \times n$

U
square $m \times m$
cols are orthonormal & linearly indep. & form basis for $\mathbb{R}^m$

Λ
$m \times n$, zero padded

VT
square $n \times n$, rows orthonormal & linearly indep. & form basis for $\mathbb{R}^n$

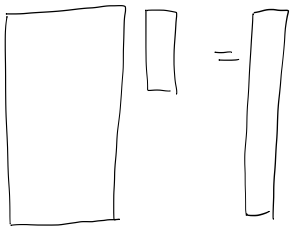
Origin of SVD?

→ generalisation of eigen-analysis to non-square matrices

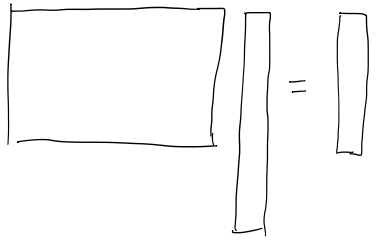
Recall that for a square matrix A, eigenvalues solve

$$Ax = \lambda x$$

However, we are interested in non-square matrices in inverse problems (& statistics etc)!



Tall / overdetermined systems
- 'classical statistics'



Wide / underdetermined systems
- 'inverse problems'
- 'nonparametric statistics'
- 'machine learning' etc

→ eigenvalues don't make sense for non-square!

Eigenvalues?

A $m \times n$

x n

y m

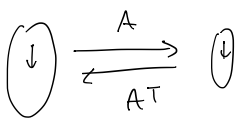
$Ax = y$

$Ax = \lambda x$ doesn't make sense

$x \in \mathbb{R}^n$, $Ax \in \mathbb{R}^m$ } live in different spaces!

Solutions?

related
- Consider different bases for each space
- Consider eigenvalues of square matrices like A^TA & AA^T:



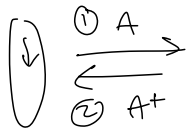
A^TA: $\mathbb{R}^n \to \mathbb{R}^n$	AA^T: $\mathbb{R}^m \to \mathbb{R}^m$

Model space: A^TA

First consider A^TA, $A: m \times n$

$A^T: n \times m$

$A^TA: n \times n$



$\mathbb{R}^n$

$A^TA: \mathbb{R}^n \to \mathbb{R}^n$

Properties of A^TA
- square: $n \times n$
- symmetric: $(A^TA)^T = A^TA$
- positive semi-definite: $x^TA^TAx = (Ax)^TAx = \|Ax\|^2 \geq 0$
- $\text{rank}(A^TA) = \text{rank}(A)$

Facts about symmetric / P.S.D matrices

1. If an $n \times n$ matrix has real entries & is symmetric then it has n distinct, real, linearly independent & orthogonal eigenvectors : basis for $\mathbb{R}^n$!

2. If an $n \times n$ matrix is symmetric & positive semi-definite then its eigenvalues are ≥ 0

3. The rank of a symmetric $n \times n$ matrix is equal to the number of non-zero (here, positive) eigenvalues

---

Imply : we can use normalised eigenvectors of A^TA as a basis for $\mathbb{R}^n$ (model space)

↳ Solve $(A^TA)v_i = \lambda_i^2 v_i$ ($\lambda_i^2 \geq 0$)

↳ get n solns $\{(\lambda_i, v_i)\}$ $i = 1, \dots n$

↳ orthonormal basis $\{v_i\}_{i=1,\dots n}$

$$v_i^T v_j = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

Data space

Before considering $AA^T : \mathbb{R}^m \to \mathbb{R}^m$

lets consider the image of our $\mathbb{R}^n$ basis vectors, ie $Av_i \in \mathbb{R}^m$

Are these orthogonal?

$$\begin{aligned} (Av_i)^T(Av_j) &= v_i^T A^T A v_j \\ &= v_i^T \lambda_j^2 v_j \qquad \text{eig.} \\ &= \lambda_j^2 \underbrace{v_i^T v_j}_{i \neq j} \qquad v \text{ orthog.} \\ &= \begin{cases} \lambda_i^2 & i = j \\ 0 & i \neq 0 \end{cases} \end{aligned}$$

Yes!

To make orthonormal, divide Av_i by $\lambda_i \neq 0$:

$$u_i = \frac{Av_i}{\lambda_i} \iff \boxed{Av_i = \lambda_i u_i}$$

for $i = 1, \dots,$ rank ($\lambda_i > 0$ for $i = 1, \dots, r$)

Properties ?

$$u_i = \frac{Av_i}{\lambda_i} \quad , \lambda_i > 0$$

Consider

$$AA^T u_i = AA^T \frac{Av_i}{\lambda_i}$$

$$= A\left(\frac{\lambda_i^2 v_i}{\lambda_i}\right) \quad [\text{eig}^{\text{n}}\ v]$$

$$= A\lambda_i v_i$$

$$= \lambda_i (Av_i) \quad [\text{def}^{\text{n}}\ u_i]$$

$$= \lambda_i^2 u_i$$

$\Rightarrow$ u_i are eigenvectors of AA^T with eigenvalues λ_i^2 !

For $\lambda_i = 0$, still define u_i as eigenvector solving $A^T A u = \lambda^2 u = 0$

Get m solutions $\{(\lambda_i, u_i)\}\ i = 1, \ldots, m$

$\{u_i\}$ basis for $\mathbb{R}^m$

Singular values & singular vectors I

Instead of $\boxed{Av = \lambda v}$ } eigenvector v, eig value λ

Consider $\boxed{Av = u\lambda}$ — scalar

matrix, vector, vector

where : $\boxed{\begin{aligned} A^T A v &= \lambda^2 v \\ AA^T u &= \lambda^2 u \end{aligned}}$ $\begin{matrix} \} i = 1, \ldots n \\ \} i = 1, \ldots m \end{matrix}$

sol^{n}s : For $\lambda_i > 0$, get triples solving all eq^{n}s,

$$\{(u_i, \lambda_i, v_i)\} \quad i = 1, \ldots, r,$$

left singular vector (data space) — singular value — right singular vector (model space)

Also get extra u, v to 'fill out' $\mathbb{R}^m, \mathbb{R}^n$ corresponding to $\lambda = 0$ eig. values of $A^T A$ &/or AA^T

Singular values & singular vectors II

In $A v_i = u_i \lambda_i$

ie $\{(u_i, \lambda_i, v_i)\}$ $i = 1, \ldots, \text{rank}$

we have:

$\{u_i\}$ are normalised eigenvectors of AA^T for $i = 1, \ldots \text{rank}$

$\{v_i\}$ are normalised eigenvectors of A^TA for $i = 1, \ldots \text{rank}$

Both sets are orthonormal

$$u_i^T u_j = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

$$v_i^T v_j = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases}$$

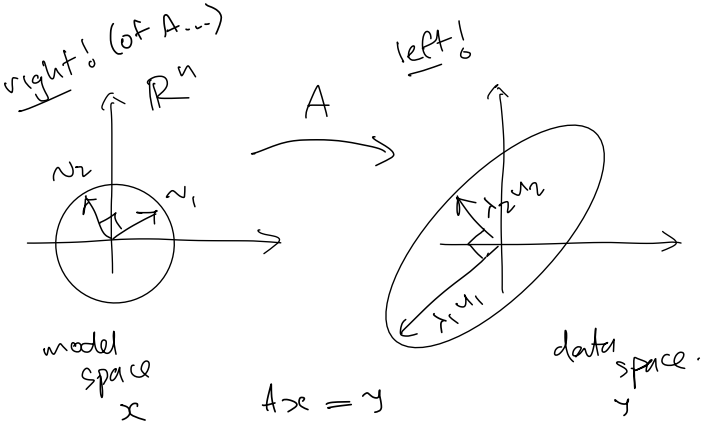
Our triplets here are defined for $\boxed{\lambda_i > 0}$ but $\{u_i\}$ for all eigenvectors of AA^T & $\{v_i\}$ for all eig. vectors span their respective spaces:

Span $\{u_i\} = \mathbb{R}^m$ for $i = 1, \ldots, m$

Span $\{v_i\} = \mathbb{R}^n$ for $i = 1, \ldots, n$

Interpretation:

$$\boxed{A v_i = \lambda_i u_i}$$



Matrix form: reduced.

A: $m \times n$

$$\boxed{Av_i = u_i \lambda_i}, \quad i = 1, \dots \text{rank}$$

$\Leftrightarrow$

$v_i \in \mathbb{R}^n$, $u_i \in \mathbb{R}^m$

$$A \begin{bmatrix} v_1 & v_2 & \dots & v_r \end{bmatrix} = \begin{bmatrix} u_1 & u_2 & \dots & u_r \end{bmatrix} \begin{bmatrix} \lambda_1 & & & [0] \\ & \lambda_2 & & \\ & & \ddots & \\ [0] & & & \lambda_r \end{bmatrix}$$

$\Leftrightarrow$

$$\boxed{AV_r = U_r \Lambda_r}, \quad V_r : n \times r, \quad U_r : m \times r, \quad \Lambda_r : r \times r$$

$\rightarrow$

full/padded:

$m \times n$

$$A \begin{bmatrix} v_1 & \dots & v_n \end{bmatrix} = \begin{bmatrix} u_1 & \dots & u_m \end{bmatrix} \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_r \end{bmatrix} \begin{bmatrix} 0 & \\ & 0 \end{bmatrix}$$

ie

$$A \begin{bmatrix} V_r & V_0 \end{bmatrix} = \begin{bmatrix} U_r & U_0 \end{bmatrix} \begin{bmatrix} \Lambda_r & 0 \\ 0 & 0 \end{bmatrix}$$

(A: $m \times n$; V_r: $n \times r$, V_0: $n \times (n-r)$; U_r: $m \times r$, U_0: $m \times (m-r)$; Λ_r: $r \times r$)

$\Leftrightarrow$

$$\boxed{AV = U\Lambda}$$

Note: 'inefficient' but $V^{-1} = V^T$, $U^{-1} = U^T$ because have full orthogonal basis

$\Rightarrow$ $AVV^T = U\Lambda V^T$

$\Rightarrow$ $\boxed{A = U\Lambda V^T}$ (full SVD)

Reduced SVD



It can be verified that

$$A = U \Lambda V^T$$ full SVD of A

expands to

$$A = U_r \Lambda_r V_r^T$$ reduced SVD of A

However, we don't have $V_r V_r^T = I$ in general (tho do have $V_r^T V_r = I$ in general) so can't so easily go from $AV_r = U_r \Lambda_r$ to $A = U_r \Lambda V_r^T$

→ Can instead show eg $AV_r V_r^T = A$ even if $V_r V_r^T \neq I$!

SVD key properties I.

$A: \mathbb{R}^n \to \mathbb{R}^m$, $m \times n$

$$A = U \Lambda V^T \quad \text{full}$$

$$A = U_r \Lambda_r V_r^T \quad \text{reduced.}$$

---

where: $U^T U = I_m = U_r^T U_r$

$V^T V = I_n = V_r^T V_r$

$UU^T = I_m$, $U_r U_r^T \neq I$ in gen.

$VV^T = I_n$, $V_r V_r^T \neq I$ in gen.

$$A^T = V \Lambda^T U^T$$

$$= V_r \Lambda_r U_r^T \quad (\Lambda_r^T = \Lambda_r)$$

$$A^T A = V \Lambda^T \Lambda V^T$$

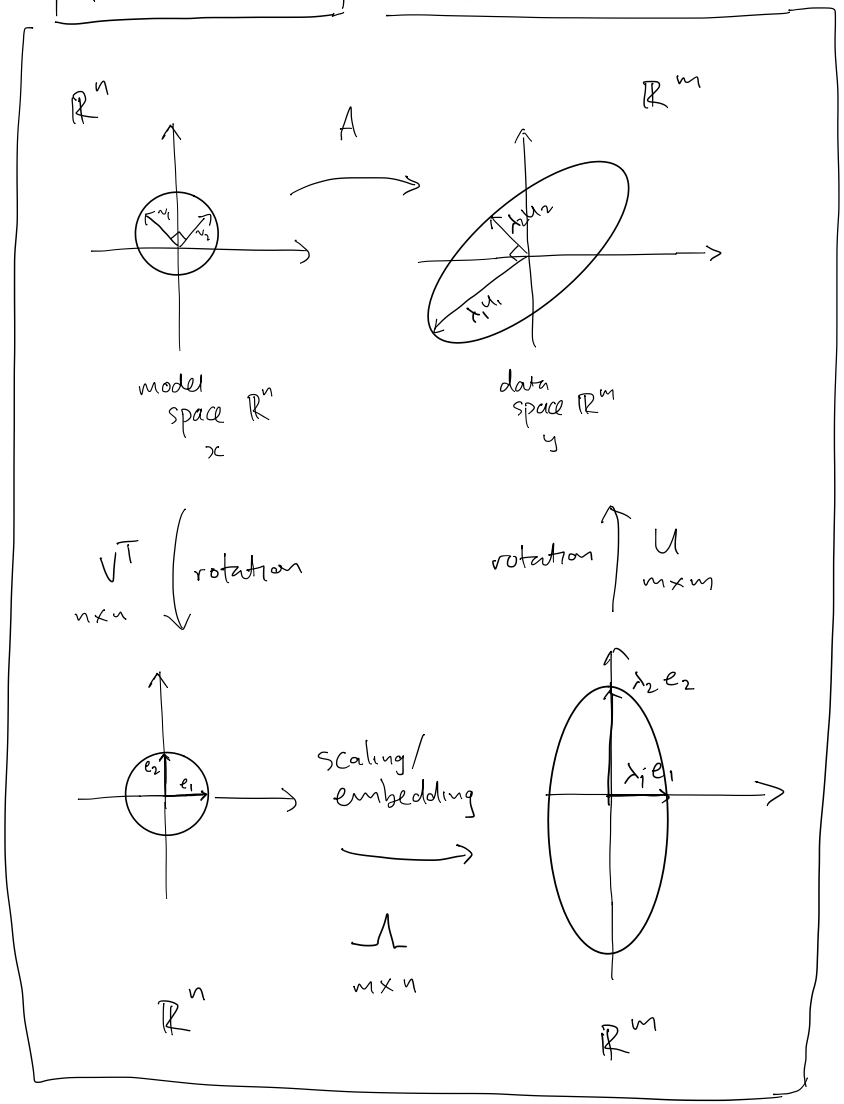
$$= V_r \Lambda_r^2 V_r^T$$

$$AA^T = U \Lambda \Lambda^T U^T$$

$$= U_r \Lambda_r^2 U_r^T$$

SVD : interpretation : U, V 'orthogonal' matrices = rotations!

$$A = U \Lambda V^T \quad \text{(full)}$$



SVD & inverses

Given:

$$A = U \Lambda V^T = U_r \Lambda_r V_r^T$$

The pseudo inverse is given by

$$A^+ = V \Lambda^+ U^T = V_r \Lambda_r^{-1} U_r^T$$

Pseudo inverse : $\boxed{A^+ = V_r \Lambda_r^{-1} U_r^T}$

- Check $AA^+A = A$?

$$(U_r \Lambda_r \underbrace{V_r^T}_{I})(V_r \Lambda_r^{-1} \underbrace{U_r^T}_{I})(U_r \Lambda_r V_r^T)$$

$$= U_r \Lambda_r \underbrace{\Lambda_r^{-1} \Lambda_r}_{I} V_r^T$$

$$= U_r \Lambda_r V_r^T = A \checkmark$$

- $(A^+A)^T = A^+A$?

$$A^+A = V_r \Lambda_r^{-1} \underbrace{U_r^T U_r}_{I} \Lambda_r V_r^T$$

$$= V_r V_r^T$$

$$A^T (A^+)^T = V_r \Lambda_r U_r^T U_r \Lambda_r^{-1} V_r^T$$

$$= V_r V_r^T \checkmark$$

etc!

Useful properties of $A = U_r \Lambda_r V_r^T$ } $V_r : n \times r$, $U_r : m \times r$

$n \times n$ { $\boxed{R_M = A^+A = V_r V_r^T}$ } projection on model space (row space of A)

$m \times m$ { $\boxed{R_D = AA^+ = U_r U_r^T}$ } projection on data space (col space of A)

$\boxed{A = AV_r V_r^T}$, since $AA^+A = AR_M$
$= AV_r V_r^T$

$\boxed{A = U_r U_r^T A}$, since $AA^+A = R_D A$
$= U_r U_r^T A$

& $\boxed{A = U_r U_r^T A V_r V_r^T}$ by above.

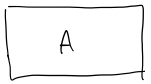
→

SVD & Resolution : Explicit cases

Now : $\boxed{R_D = U_r U_r^T}$ $m \times m$ { U_r $m \times r$ (r nvectors); U_r^T $r \times m$

$\boxed{R_M = V_r V_r^T}$ $n \times n$ { V_r $n \times r$ (r nvectors); V_r^T $r \times n$

- If rank $r = m < n$

  $\Rightarrow R_D = I_m$ 

(recover data exactly)
(models are 'reduced')

  But $R_M \neq I_n$

  Though $R_M^2 = V_r V_r^T V_r V_r^T = V_r V_r^T = R_M$

  $\Rightarrow R_M$ is model projection operator

- If rank $r = n < m$

  $R_D \neq I_m$

  $R_M = I_n$ 

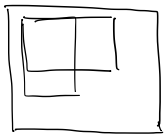
(data are 'reduced')
(models recovered exactly.)

  though $R_D = U_r U_r^T U_r U_r^T = U_r U_r^T = R_D$

  $\Rightarrow R_D$ is data projection

- If rank $r < m$ & n

  $R_D \neq I_m$

  $R_M \neq I_n$ 

  $\Rightarrow$ Both are projection operators.

  ($U_r^T U_r = I$, $V_r^T V_r = I$ still)

SVD : Big picture

Key advantage : explict calculation

- inverses (left, right, psendo)
- model / data resolution operators

coming up:
- stability / instability depending on singular values
- stablised approximations via truncation
- effect of Tikhonov (etc) regularisation on singular values

Disadvantage : though some intuitions transfer to nonlinear, essentially a linear concept.

↳ But see rank factorisation & mono-epi factorisation

↳ can apply to eg linearisation

So.... regularisation!

→ singular values may be positive but effectively zero (machine tol. etc)

⇒ cause: **effective rank p < rank r**

→ small singular values cause instability

Key: inverse leads to

dividing by small λ_i values

SVD as basis expansion:

$$A^\dagger = V_r \Lambda_r^{-1} U_r^T = \sum_{i=1}^{r} v_i \frac{1}{\lambda_i} u_i^T$$

& $x^\dagger = A^\dagger y = \left(\sum_{i=1}^{r} v_i \frac{1}{\lambda_i} u_i^T \right) y$

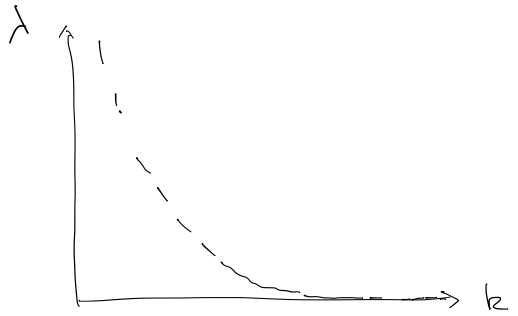
$$= \sum_{i=1}^{r} \left(\frac{u_i^T y}{\lambda_i} \right) v_i$$

coeff. ↓ Large for $\lambda_i \to 0$

basis vector in model space

Spectrum

Plot of singular values in decreasing order:



Key: **ill-posed**

→ no clear gap
→ decreases to zero

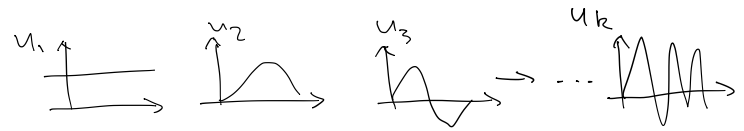
} rank is hard to define

[cf: rank deficient: clear gap. $A^\dagger$ OK then?]

Also: singular **vectors** typically **'oscillate'** more (sign changes in elements) for smaller values for ill-posed problems

↳ Like Fourier bases (see example)

note: ✻ sign changes



Truncate ~ low pass filter, but basis determined by A.

Truncated SVD!

$$A_k^+ = V_k \Lambda_k^{-1} U_k^T$$ } Truncated SVD inverse

→ remove small singular values/ oscillatory basis vectors

→ a form of regularisation/ stabilisation

Resolution?

$$R_M = A_k^+ A$$

$$R_D = A A_k^+$$

$$A \rightleftarrows A_k^+$$

Stability

consider $x^+ = A^+ y$

& $x^{+\prime} = A^+ y'$

for small data perturbation

$$\| y - y' \|_2 < \delta$$

Then $x^+ - x^{+\prime} = A^+ (y - y')$

& $\| x^+ - x^{+\prime} \|_2 \leq \| A^+ \|_2 \| y - y' \|_2$

where $\| A \|_2 := \max_{\|x\|_2 = 1} \| Ax \|_2 = \lambda_1$

$= $ largest singular value

leads to (with other details---)

$$\frac{\| x^+ - x^{+\prime} \|_2}{\| x^+ \|_2} \leq \frac{\lambda_1}{\lambda_r} \frac{\| y - y' \|_2}{\| y \|_2}$$

λ_1: largest singular value

λ_r: smallest singular value.

Stability : Key point

Stability (continuity modulus) of A^+ governed by

$$\text{cond}(A^+) = \frac{\lambda_1}{\lambda_r}$$

(condition number)

Key trade-off :

truncate singular value expansion

↳ more stable (less 'variance')

↳ biased (model resolution less like identity

→ favour particular models)

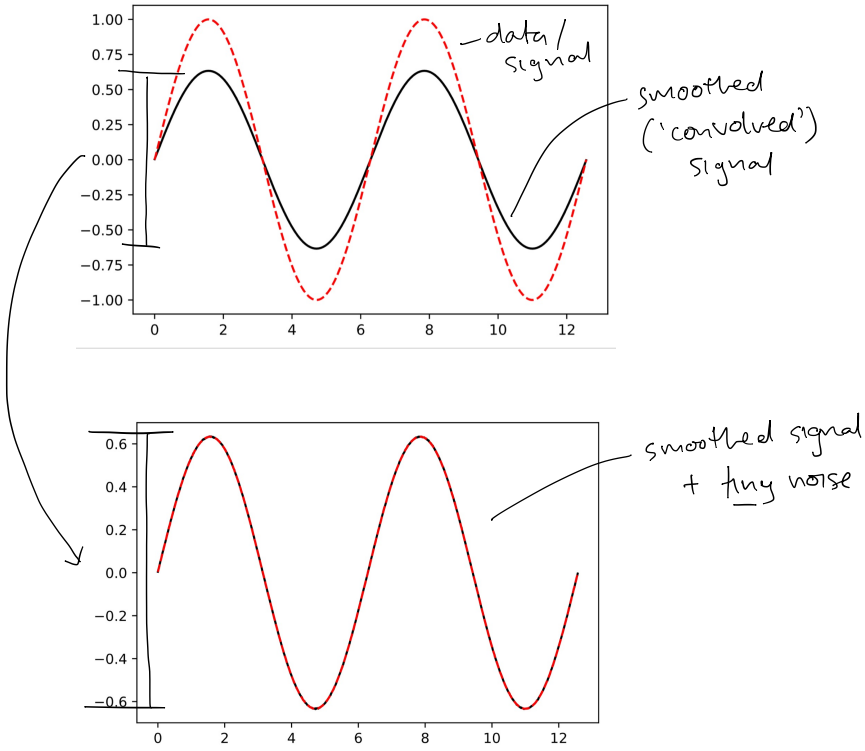
(stats : Bias - Variance tradeoff)

Example

Return to deconvolution example from L1.

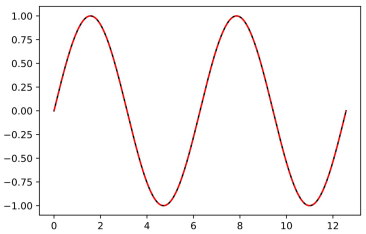
(convolution ≈ window averaging
deconvolution ≈ --undoing ↑ !)

Ax
$+$
ϵ



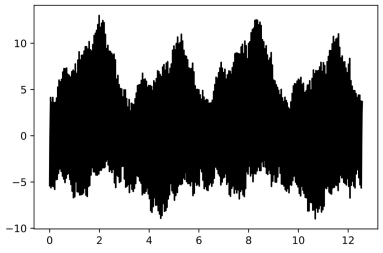
Example

Deconvolution no noise



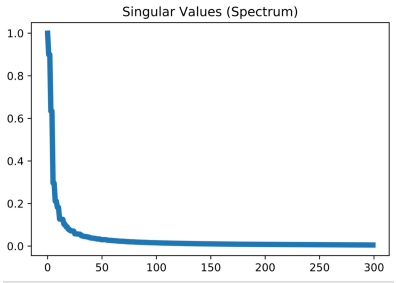
yay!

Deconvolution with noise

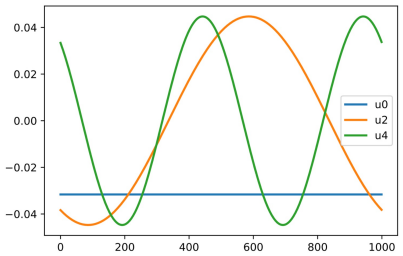


noo!

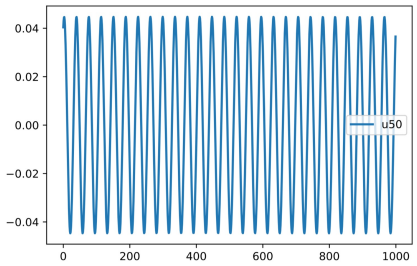
SVD: spectrum



u vectors (v similarly)



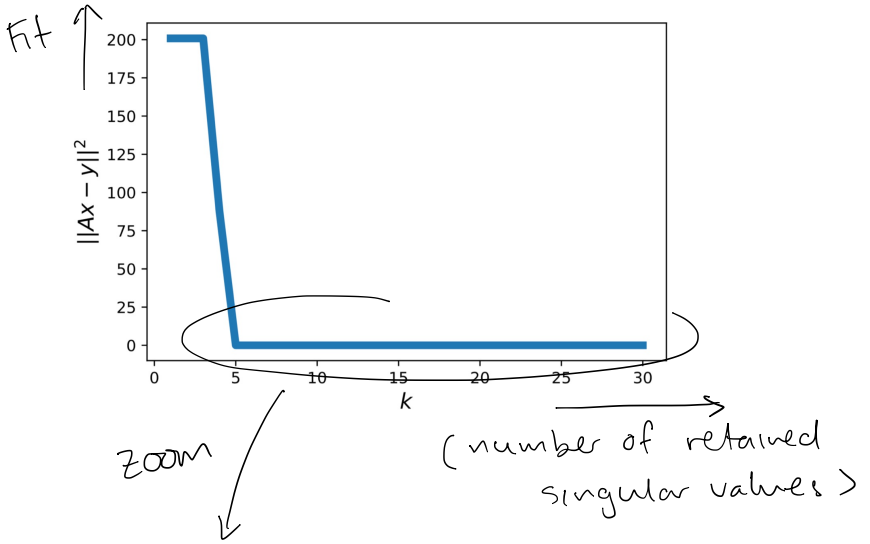
u vector for small λ_i:

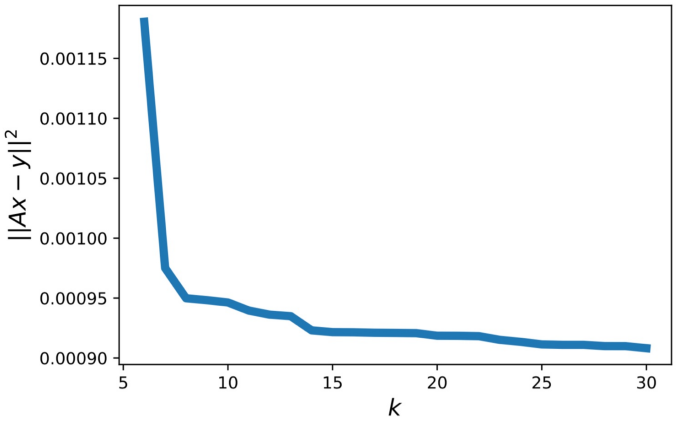


Think: Fourier components.

(but 'tailored' to A)

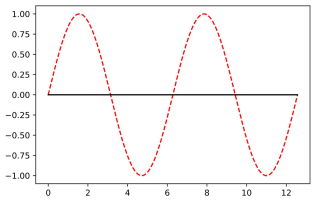
Pareto (trade-off) curve:





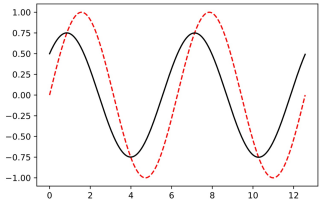
Solutions as depending on k (number retained singular values)

Stable/ Biased

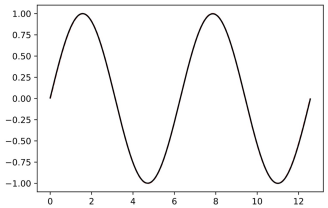


$k = 1$

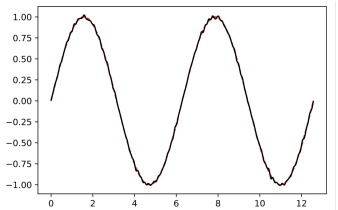
— recovered
--- true



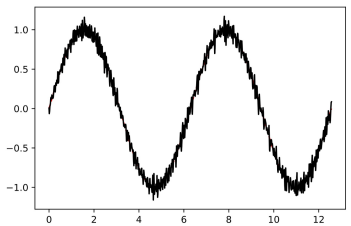
$k = 4$



$k = 5$ ← sweet spot



$k = 100$



$k = 300$

(this is a very smooth problem — typically much worse, faster)

Unstable/ Unbiased

Choosing truncation ?

- Pareto : Smallest number of singular values giving adequate fit, beyond which 'flattens'
  - ↳ 'L-curve' -like
- Picard condition

Consider series rep. of SVD:

$$x = \sum_{i=1}^{r} \left[\left(\frac{u_i^T y}{\lambda_i} \right) v_i \right]$$

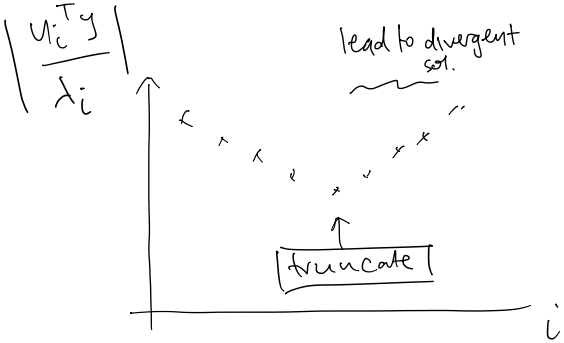
($\frac{u_i^T y}{\lambda_i}$: coeff. v_i: model basis)

series converge ? ⇒ need coeff → 0 !

plot $\left| \frac{u_i^T y}{\lambda_i} \right|$ vs i [exercise : do for convolution!]

Expect : $u_i^T y$ decay faster than λ_i at first, then λ_i faster & ratio starts to increase

→ truncate here



Note : relative decay 's what matters, not abs. magnitude

Note:

TSVD & optimality (Eckart-Young)

- The optimal rank k approx. to a matrix A in L_2 norm is given by the rank k truncation of the SVD

$$\Rightarrow \operatorname{argmin}_{\{\hat{A} \mid \hat{A} \text{ rank } k\}} \| A - \hat{A} \|_2 = U_k \Lambda_k V_k^T$$

$= \text{rank } k \text{ SVD}$

$= \text{rank } k \text{ truncation of SVD}$

Tikhonov & SVD

Finally, lets return to Tikhonov regularisation & see if SVD can help understand.

Zeroth order: Normal eq^{n}s

(instead of λ to simplify)

$$\boxed{(A^T A + \alpha^2 I)x = A^T y}$$

Consider SVD of original & augmented..

→

... Tikhonov & SVD ... Spectral filtering.

Can show

$$x = \sum_{i=1}^{r} \left(\frac{u_i^T y}{\lambda_i} \right) v_i \qquad \text{(SVD)}$$

coeff. · model basis

becomes for Tikhonov regularised:

$$x_\alpha = \sum_{i=1}^{r} \left(f_i \cdot \frac{u_i^T y}{\lambda_i} \right) v_i$$

} general form of 'spectral filtering' methods

where

$$f_i = \frac{\lambda_i^2}{\lambda_i^2 + \alpha^2}$$

are the **filter factors**

Note:
- $\alpha = 0 \Rightarrow f_i = 1$
- $\lambda_i << \alpha \Rightarrow f_i \to \left(\frac{\lambda_i}{\alpha}\right)^2 \to 0$
- $\lambda_i >> \alpha \Rightarrow f_i \to 1$

$\Rightarrow$ Tikhonov regularisation implements (continuous version of) truncated SVD !

→ diff. f_i give diff. 'spectral regularisation': Eg TSVD: $f_i = \begin{cases} 1 & i \le k \\ 0 & i > k \end{cases}$

Exercises

1. Compare & contrast the standard eigenvalue problem for a matrix A with the problem giving the singular values/vectors of A

2. Write down the SVD in the forms
   - full
   - reduced
   - series expansion

3. Write down the expression for A^+ given the SVD of A.

4. Verify A^+ is a left/right inverse when one exists, using the SVD representation

5. Write down expressions for R_D & R_m resolution matrices in terms of SVD components

Hints

4. Left inverse exists for tall A with linearly independent cols.

Consider $A = U_r \Lambda_r V_r^T$

$A^+ = V_r \Lambda_r^{-1} U_r^T$

$\Rightarrow \quad A^+ A = V_r V_r^T$

But A: m × $n = r$ } linearly indep. cols.

full V = basis vectors for $\mathbb{R}^n$ (model space)

= basis for $\mathbb{R}^r$ ($r = n$ as lin. indep)

$= V_r$

full V always satisfies $VV^T = I$ (orthog. basis)

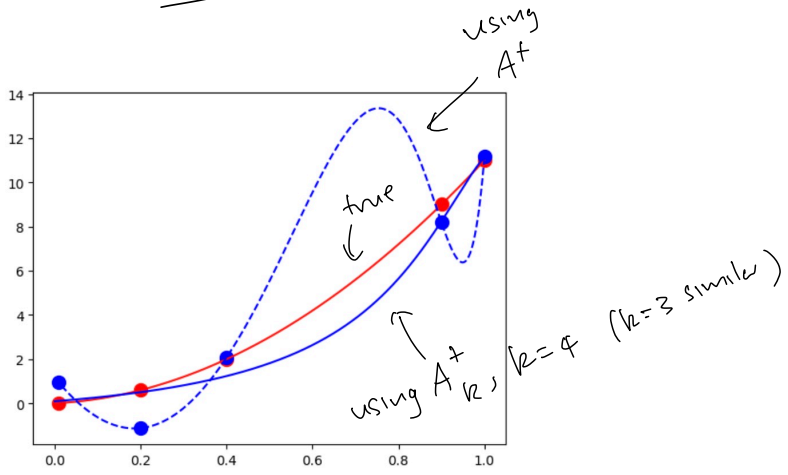
Here $V_r = V \Rightarrow A^+ A = V_r V_r^T = VV^T = I$

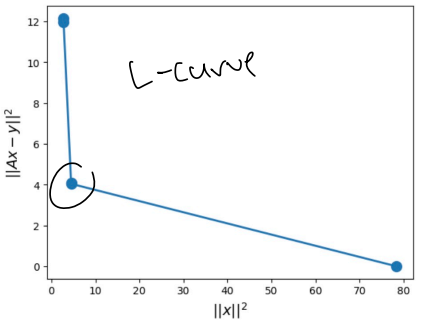
$\Rightarrow A^+$ is left inverse

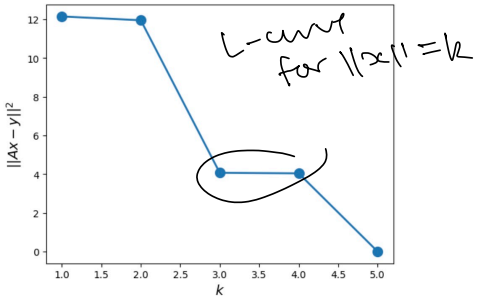
ie left inverse $\Leftrightarrow V_r = V$

$\Leftrightarrow$ linearly indep. cols

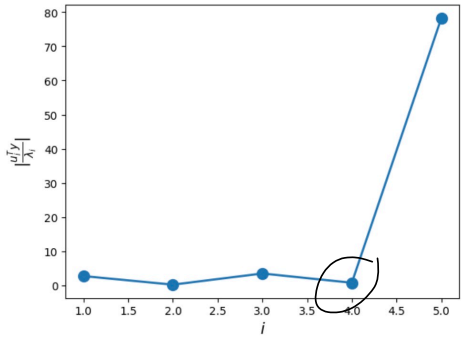
SVD: regression example cont'd (x^{19})







Picard:



Model resolution

